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In this lab, I will use my Kali Linux attacker VM to perform network reconnaissance and scan other hosts within my lab environment. I will then collect and analyse the logs generated by this activity using Splunk to identify evidence of network scanning and reconnaissance.

Nmap Port Scan Against Ubuntu Endpoint

Kali (Attacker) → Nmap scan → Ubuntu 192.168.50.30 (Target) → logs → Splunk (SIEM)

Network host discovery.

An Nmap ping scan was done against SOC-LAB subnet to identify active hosts.

```
(joanna@kali)-[~]  
$ sudo nmap -sn 192.168.50.0/24  
[sudo] password for joanna:  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-26 13:32 BST  
Nmap scan report for 192.168.50.10  
Host is up (0.00025s latency).  
MAC Address: BC:24:11:EC:DD:12 (Unknown)  
Nmap scan report for 192.168.50.30  
Host is up (0.00033s latency).  
MAC Address: BC:24:11:F8:7A:D5 (Unknown)  
Nmap scan report for 192.168.50.40  
Host is up (0.00028s latency).  
MAC Address: BC:24:11:1B:56:FD (Unknown)  
Nmap scan report for 192.168.50.100  
Host is up.  
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.04 seconds
```

Scan identified four active systems in the 192.168.50.0/24 network. Host Ubuntu was chosen as the initial target for further reconnaissance.

Port scanning Ubuntu endpoint

```
(joanna@kali)-[~]
$ sudo nmap -sS -sV 192.168.50.30
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-08-26 13:36 BST
Nmap scan report for 192.168.50.30
Host is up (0.000040s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
222/tcp   open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; protocol 2.0)
MAC Address: BC:24:11:F8:7A:D5 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

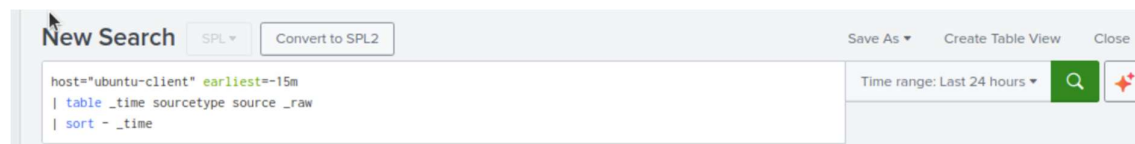
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.67 seconds

(joanna@kali)-[~]
$
```

The TCP SYN and service enumeration scan identified one open port on the Ubuntu endpoint. Port 222/TCP was open and running OpenSSH 9.6p1. The remaining 999 scanned TCP ports were closed.

Splunk Log Analysis

In this step, I used Splunk to investigate whether the activity performed in previous steps generated observable events on the target Ubuntu. In my search, I searched for the logs from last 15 minutes.



Initial inspection showed an authentication log event that was generated after Nmap service scan was performed by an IP address 192.168.50.100. This address corresponds to Kali attacker VM.

2026-08-26 13:40:11.586	syslog	/var/log/ syslog	2026-08-26T13:40:11.586701+01:00 ubuntu-client systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-08-26 13:40:11.586	syslog	/var/log/ syslog	2026-08-26T13:40:11.586514+01:00 ubuntu-client systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-08-26 13:40:11.075	syslog	/var/log/ syslog	2026-08-26T13:40:11.075148+01:00 ubuntu-client systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-08-26 13:38:47.535	syslog	/var/log/ syslog	2026-08-26T13:38:47.535160+01:00 ubuntu-client systemd[1]: NetworkManager-dispatcher.service: Deactivated successfully.
2026-08-26 13:38:37.525	syslog	/var/log/ syslog	2026-08-26T13:38:37.525178+01:00 ubuntu-client systemd[1]: Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
2026-08-26 13:38:37.524	syslog	/var/log/ syslog	2026-08-26T13:38:37.524753+01:00 ubuntu-client dbus-daemon[754]: [system] Successfully activated service 'org.freedesktop.nm_dispatcher'
2026-08-26 13:38:37.516	syslog	/var/log/ syslog	2026-08-26T13:38:37.516148+01:00 ubuntu-client systemd[1]: Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
2026-08-26 13:38:37.492	syslog	/var/log/ syslog	2026-08-26T13:38:37.492955+01:00 ubuntu-client dbus-daemon[754]: [system] Activating via systemd: service name='org.freedesktop.nm_dispatcher' unit='dbus-org.freedesktop.nm_dispatcher.service' requested by ':1.14' (uid=0 pid=878 comm="/usr/sbin/NetworkManager --no-daemon" label="unconfined")
2026-08-26 13:38:37.491	syslog	/var/log/ syslog	2026-08-26T13:38:37.491706+01:00 ubuntu-client NetworkManager[878]: <info> [1787747917.4912] dhcp4 (ens18): state changed new lease, address=192.168.5.109
2026-08-26 13:36:53.283	linux_secure	/var/log/ auth.log	2026-08-26T13:36:53.283804+01:00 ubuntu-client sshd[9988]: Connection closed by 192.168.50.100 port 60428
2026-08-26 13:35:01.625	linux_secure	/var/log/ auth.log	2026-08-26T13:35:01.625116+01:00 ubuntu-client CRON[9953]: pam_unix(cron:session): session closed for user root
2026-08-26 13:35:01.625	syslog	/var/log/ syslog	2026-08-26T13:35:01.625041+01:00 ubuntu-client CRON[9955]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)

To determine whether the reconnaissance activity could be associated with Kali VM, I filtered the logs and added Kali IP address (192.168.50.100).

New Search

SPL

Convert to SPL2

Save As

Create Table View

Close

host="ubuntu-client" *192.168.50.100"

| table _time sourcetype source _raw

| sort - _time

Time range: Last 24 hours

1 event (8/25/26 1:00:00.000 PM to 8/26/26 1:50:32.000 PM)

No Event Sampling

Job

Smart Mode

Events

Patterns

Statistics (1)

Visualization

Show: 20 Per Page

Format

Preview: On

As shown above, splunk returned a linux_secure event from /var/log/auth.log at 13:36:53, showing that the SSH service on the Ubuntu endpoint received and closed a connection originating from 192.168.50.100. This occurred during the Nmap service enumeration scan performed at approximately 13:36.

The event provides host-level evidence that the Kali VM interacted with the exposed SSH service during reconnaissance. However, the available auth.log and syslog data does not provide visibility of the complete SYN port scan across the other TCP ports.

Improving network visibility

To improve network visibility, I configured firewall logging on the Ubuntu endpoint so that additional network activity could be collected and analysed in Splunk.

Before enabling UFW, I added an allow rule for TCP port 222 to ensure that the SSH service remained accessible. UFW logging was then configured at medium level and the firewall was enabled.

```
soc_lab@ubuntu-client:~$ sudo ufw status verbose
[sudo] password for soc_lab:
Status: inactive
soc_lab@ubuntu-client:~$ sudo ufw allow 222/tcp
Rules updated
Rules updated (v6)
soc_lab@ubuntu-client:~$ sudo ufw logging medium
Logging enabled
soc_lab@ubuntu-client:~$ sudo ufw enable
Firewall is active and enabled on system startup
soc_lab@ubuntu-client:~$ sudo ufw status verbose
Status: active
Logging: on (medium)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From
--	-----	----
1514/tcp	ALLOW IN	Anywhere
22/tcp	ALLOW IN	Anywhere
222/tcp	ALLOW IN	Anywhere
1514/tcp (v6)	ALLOW IN	Anywhere (v6)
22/tcp (v6)	ALLOW IN	Anywhere (v6)
222/tcp (v6)	ALLOW IN	Anywhere (v6)

Repeating the Nmap Scan

After enabling firewall logging, I repeated the same Nmap SYN and service enumeration scan from the Kali attacker VM against the Ubuntu endpoint.

The purpose of repeating the same scan was to determine whether the additional firewall logging provided better visibility of the reconnaissance activity.

Detecting the Port Scan in Splunk

After repeating the scan, I searched Splunk for UFW events originating from the Kali attacker VM.

The screenshot shows a Splunk search interface with the following details:

- Search Bar:** Index=main host="ubuntu-client" "UFW BLOCK" "SRC=192.168.50.100"
- Results:** 10 events (07/09/2026 09:25:00.000 to 07/09/2026 13:25:20.000)
- Fields List:**
 - SELECTED FIELDS:** host, source, sourcetype
 - INTERESTING FIELDS:** date_hour, date_mday, date_minute, date_month, date_second, date_wday, date_year, dest, DPT, dvc, eventtype, ID, IN, index
- Event Table:**

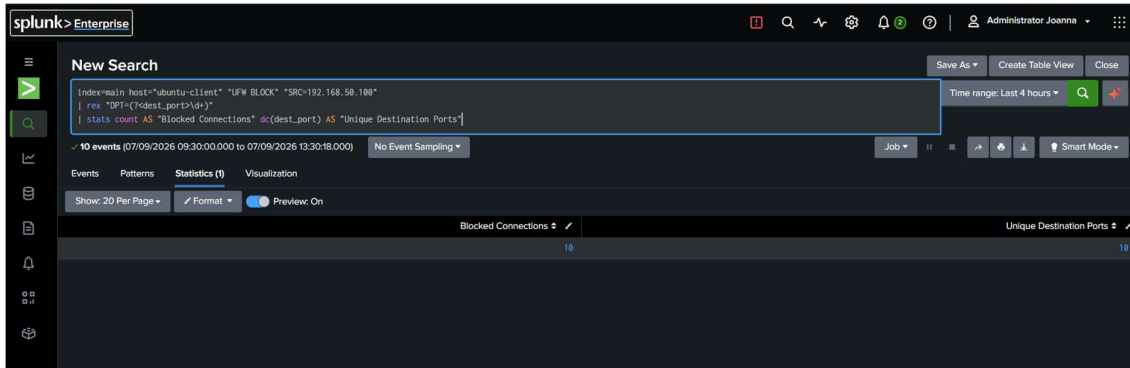
i	Time	Event
>	07/09/2026 12:07:44.919	2826-09-07T12:07:44.919651+01:00 ubuntu-client kernel: [UFW BLOCK] IN=ens19 OUT= MAC=bc:24:11:f8:7a:d5:bc:24:11:a5:d5:b0:08:00 SRC=192.168.50.100 DST=192.168.50.30 UEN=44 TOS=0x00 PREC=0x00 TTL=57 ID=33428 PROTO=TCP SPT=37137 DPT=65389 WINDOW=1024 RES=0x00 SYN URG=0 host = ubuntu-client source = /var/log/syslog sourcetype = syslog
>	07/09/2026 12:07:42.214	2826-09-07T12:07:42.214800+01:00 ubuntu-client kernel: [UFW BLOCK] IN=ens19 OUT= MAC=bc:24:11:f8:7a:d5:bc:24:11:a5:d5:b0:08:00 SRC=192.168.50.100 DST=192.168.50.30 UEN=44 TOS=0x00 PREC=0x00 TTL=37 ID=39265 PROTO=TCP SPT=37137 DPT=111 WINDOW=1024 RES=0x00 SYN URG=0 host = ubuntu-client source = /var/log/syslog sourcetype = syslog
>	07/09/2026 12:07:42.214	2826-09-07T12:07:42.214798+01:00 ubuntu-client kernel: [UFW BLOCK] IN=ens19 OUT= MAC=bc:24:11:f8:7a:d5:bc:24:11:a5:d5:b0:08:00 SRC=192.168.50.100 DST=192.168.50.30 UEN=44 TOS=0x00 PREC=0x00 TTL=53 ID=58435 PROTO=TCP SPT=37137 DPT=1723 WINDOW=1024 RES=0x00 SYN URG=0 host = ubuntu-client source = /var/log/syslog sourcetype = syslog
>	07/09/2026 12:07:42.214	2826-09-07T12:07:42.214768+01:00 ubuntu-client kernel: [UFW BLOCK] IN=ens19 OUT= MAC=bc:24:11:f8:7a:d5:bc:24:11:a5:d5:b0:08:00 SRC=192.168.50.100 DST=192.168.50.30 UEN=44 TOS=0x00 PREC=0x00 TTL=56 ID=61908 PROTO=TCP SPT=37137 DPT=110 WINDOW=1024 RES=0x00 SYN URG=0 host = ubuntu-client source = /var/log/syslog sourcetype = syslog
>	07/09/2026 12:07:42.214	2826-09-07T12:07:42.214756+01:00 ubuntu-client kernel: [UFW BLOCK] IN=ens19 OUT= MAC=bc:24:11:f8:7a:d5:bc:24:11:a5:d5:b0:08:00 SRC=192.168.50.100 DST=192.168.50.30 UEN=44 TOS=0x00 PREC=0x00 TTL=46 ID=35582 PROTO=TCP SPT=37137 DPT=1720 WINDOW=1024 RES=0x00 SYN URG=0 host = ubuntu-client source = /var/log/syslog sourcetype = syslog
>	07/09/2026 12:07:42.214	2826-09-07T12:07:42.214755+01:00 ubuntu-client kernel: [UFW BLOCK] IN=ens19 OUT= MAC=bc:24:11:f8:7a:d5:bc:24:11:a5:d5:b0:08:00 SRC=192.168.50.100 DST=192.168.50.30 UEN=44 TOS=0x00 PREC=0x00 TTL=51 ID=39256 PROTO=TCP SPT=37137 DPT=995 WINDOW=1024 RES=0x00 SYN URG=0 host = ubuntu-client source = /var/log/syslog sourcetype = syslog

Splunk returned multiple UFW BLOCK events generated during the scan. The events show TCP SYN packets from the Kali VM (192.168.50.100) targeting multiple destination ports on the Ubuntu endpoint (192.168.50.30).

Unlike the initial authentication log analysis, the firewall logs provide visibility of connection attempts to ports that were closed. This provides much clearer evidence of the Nmap port scanning activity.

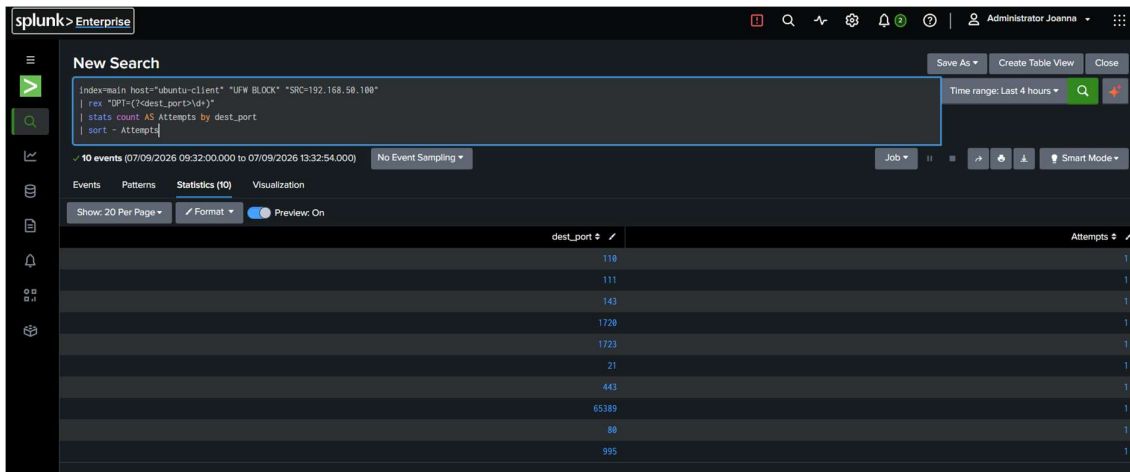
I then extracted the destination port from the UFW events and counted the number of blocked connections and unique destination ports observed in Splunk. The search identified 10 blocked connections across 10 different destination ports.

Although the Nmap scan covered 1,000 TCP ports, only a subset of the blocked packets was recorded in the available UFW logs. Despite this, the rapid connection attempts from a single source IP to multiple destination ports provide clear evidence of port scanning activity.



I then extracted the destination port from the firewall events to identify which ports were observed during the scan. Splunk recorded connection attempts to 10 different destination ports, including ports 21, 80, 110, 143 and 443.

The attempts occurred within the same short time period and originated from the Kali VM, providing further evidence of network reconnaissance against multiple services.



Port Scan Detection

After confirming that the Nmap activity was visible in the firewall logs, I created a Splunk detection to identify potential port scanning behaviour. Instead of searching specifically for the Kali IP address, the detection was designed to identify any source host attempting to connect to multiple destination ports within a short period of time.

The screenshot shows the Splunk Enterprise interface with a search for firewall events. The search results table displays data for a specific time window on 2026-09-07.

_time	src_ip	dest_ip	attempts	unique_ports	ports
2026-09-07 12:07:00	192.168.50.100	192.168.50.30	10	10	110 111 143 1728 1723 21 443 65389 88 995

The final detection grouped the firewall events into one-minute time windows and identified source hosts attempting to connect to five or more unique destination ports within that period.

The detection successfully identified the Kali VM (192.168.50.100) scanning the Ubuntu endpoint (192.168.50.30). Splunk recorded 10 connection attempts to 10 unique destination ports within the same one-minute window.

The source IP was not specified in the search, meaning that the detection identifies potential port scanning based on behaviour rather than a known attacker address. A threshold of five unique ports was used for this lab and would require further tuning in a production environment to reduce false positives.

Automated Alert

To automate the detection, I converted the SPL search into a scheduled Splunk alert. The alert runs every five minutes and analyses the previous five minutes of firewall activity. It triggers when the detection search returns one or more results.

Save As Alert

Settings

Title

Potential Port Scan Detected

Description

Detects a source host attempting connections to five or more unique TCP destination ports within a one-minute window.

Permissions

Private

Shared in App

Alert type

Scheduled

Real-time

Run on Cron Schedule

Time Range

Last 5 minutes

Cron Expression

*/5 *

e.g. 00 18 *** (every day at 6PM).

Learn More

Expires

24

hour(s)

Trigger Conditions

Trigger alert when

Number of Results

is greater than

0

Trigger

Once

For each result

Throttle

Trigger Actions

+ Add Actions

When triggered

Add to Triggered Alerts

Remove

Severity

Medium

Cancel

Save

I then repeated the Nmap scan from the Kali VM. Splunk successfully triggered the Potential Port Scan Detected alert automatically, confirming that the detection can identify new port scanning activity without requiring a manual search.

```
joanna@kali:~$ sudo nmap -SS -sV 192.168.50.30
[sudo] password for joanna:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-07 13:53 BST
Nmap scan report for 192.168.50.30
Host is up (0.00033s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    closed ssh
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.18 (Ubuntu Linux; proto
col 2.0)
MAC Address: BC:24:11:F8:7A:D5 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/Submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.19 seconds
joanna@kali:~$
```

splunk> Enterprise

Triggered Alerts

Q. Filter

App All apps

Owner All owners

Severity All severity

Alert name All alerts

Showing 1 - 1 of 1 results

Show: 20 per page

1

of 1 pages

Time	Alert name	App	Type	Severity	Mode	Actions
2026-09-07 13:55:02 BST	Potential Port Scan Detected	search	Scheduled	Medium	Digest	View Results Edit Alert Delete

Summary

This lab showed how an Nmap port scan can be detected using UFW logs and Splunk. After enabling firewall logging, I was able to see connections to multiple ports and create a Splunk search to detect this behaviour.

Finally, I created an automated alert and confirmed that it successfully triggered when the Nmap scan was repeated.